V001 — STOP VALVE (GLOBE)

id	V001
component_name	V001 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	V001
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	50
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA546_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA546_IN, PIPE_FA546_OUT
relationships.related_component_ids	

V002 — BALL VALVE

id	V002
component_name	V002 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	V002
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA582_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA582_IN, PIPE_FA582_OUT
relationships.related_component_ids	

V003 — BALL VALVE

id	V003
component_name	V003 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	V003
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	80
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	24.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA572_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA572_IN, PIPE_FA572_OUT
relationships.related_component_ids	

V004 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	V004
component_name	V004 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	V004
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA512_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA512_IN, PIPE_FA512_OUT
relationships.related_component_ids	

V005 — GATE VALVE

id	V005
component_name	V005 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	V005
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	100
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	52.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA542_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA542_IN, PIPE_FA542_OUT
relationships.related_component_ids	

T-001 — FUEL OIL STORAGE TANK

id	T-001
component_name	T-001 - FUEL OIL STORAGE TANK
component_type	Tank
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Oil Storage Area
identification.line_number	T-001
specifications.tank_type	Vertical cylindrical storage tank
specifications.capacity	5000
specifications.capacity_unit	Litres
specifications.material	Carbon steel (mock value)
specifications.orientation	Vertical
specifications.design_pressure	Atmospheric (mock value)
specifications.operating_pressure	0.1
specifications.pressure_unit	bar
specifications.design_temperature	60
specifications.temperature_unit	°C
specifications.fluid_type	Fuel Oil (F.O.)
specifications.inlet_connection	DN80 flanged (mock value)
specifications.outlet_connection	DN80 flanged (mock value)
specifications.vent_connection	DN50 (mock value)
specifications.drain_connection	DN25 (mock value)
specifications.level_measurement	Level transmitter
specifications.high_level_alarm	90%
specifications.low_level_alarm	15%
usecase	Stores Fuel Oil (F.O.) for the filling and transfer system. All tank specifications shown here are illustrative mock values.
troubleshooting.fault_code	T001_F001
troubleshooting.failure_condition	Abnormal tank level indication / suspected leakage (mock scenario)
troubleshooting.symptoms	Unexpected level variation, high/low level alarm, visible leakage, or abnormal filling duration.
troubleshooting.possible_causes	Level sensor fault, loose instrument connection, inlet/outlet valve issue, tank leakage, or incorrect level calibration.
troubleshooting.diagnostic_steps	Compare local level indication with transmitter reading, inspect visible tank surfaces and connections, review alarm history, and verify associated line-up.
troubleshooting.immediate_action	Inform the responsible engineer, follow vessel procedures, stop transfer if leakage or unsafe level is suspected, and do not enter the tank.
troubleshooting.operational_impact	Reduced storage availability, transfer interruption, or potential fuel loss.
troubleshooting.verification_steps	Confirm stable level indication, inspect for leakage, verify alarm operation, and obtain engineer acceptance after corrective work.
troubleshooting.prevention_measures	Inspect tank shell and connections, test level instruments and alarms, and follow the approved inspection schedule.
troubleshooting.related_component_ids	V001, V002 (mock associations)
simulation.current_status	STANDBY
simulation.current_level	0
simulation.level_percentage	0
simulation.capacity	5000
simulation.remaining_capacity	5000
simulation.inlet_status	CLOSED
simulation.outlet_status	CLOSED
simulation.inlet_flow_rate	0
simulation.outlet_flow_rate	0
simulation.pressure	0.1
simulation.temperature	25
simulation.high_level_alarm	false
simulation.low_level_alarm	true
simulation.leakage_status	false
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_T001_IN, PIPE_T001_OUT (mock IDs)
relationships.related_component_ids	V001, V002 (mock IDs)

T-002 — FUEL OIL SERVICE TANK

id	T-002
component_name	T-002 - FUEL OIL SERVICE TANK
component_type	Tank
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Oil Service Area
identification.line_number	T-002
specifications.tank_type	Rectangular service tank
specifications.capacity	3000
specifications.capacity_unit	Litres
specifications.material	Carbon steel (mock value)
specifications.orientation	Horizontal / rectangular
specifications.design_pressure	Atmospheric (mock value)
specifications.operating_pressure	0.1
specifications.pressure_unit	bar
specifications.design_temperature	60
specifications.temperature_unit	°C
specifications.fluid_type	Fuel Oil (F.O.)
specifications.inlet_connection	DN65 flanged (mock value)
specifications.outlet_connection	DN65 flanged (mock value)
specifications.vent_connection	DN40 (mock value)
specifications.drain_connection	DN25 (mock value)
specifications.level_measurement	Level transmitter with local gauge
specifications.high_level_alarm	90%
specifications.low_level_alarm	15%
usecase	Provides a service/storage supply of Fuel Oil (F.O.) to the transfer system. All tank specifications shown here are illustrative mock values.
troubleshooting.fault_code	T002_F001
troubleshooting.failure_condition	Low tank level / restricted outlet flow (mock scenario)
troubleshooting.symptoms	Low-level alarm, reduced outlet flow, unstable downstream pressure, or unexpected level drop.
troubleshooting.possible_causes	Low tank inventory, outlet strainer restriction, closed or partially open outlet valve, level sensor error, or leakage.
troubleshooting.diagnostic_steps	Verify local and remote level readings, inspect outlet line and strainer under approved procedure, confirm valve position, and compare inlet/outlet flow readings.
troubleshooting.immediate_action	Notify the responsible engineer, verify available fuel quantity, and follow approved transfer/isolation procedures. Do not open or enter the tank.
troubleshooting.operational_impact	Reduced fuel supply, transfer interruption, or downstream equipment limitation.
troubleshooting.verification_steps	Confirm level indication, restore normal outlet flow, check pressure stability, and obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Monitor tank level, inspect outlet connections and strainers, test alarms, and maintain instruments per maker recommendations.
troubleshooting.related_component_ids	V003, V004, V005 (mock associations)
simulation.current_status	STANDBY
simulation.current_level	0
simulation.level_percentage	0
simulation.capacity	3000
simulation.remaining_capacity	3000
simulation.inlet_status	CLOSED
simulation.outlet_status	CLOSED
simulation.inlet_flow_rate	0
simulation.outlet_flow_rate	0
simulation.pressure	0.1
simulation.temperature	25
simulation.high_level_alarm	false
simulation.low_level_alarm	true
simulation.leakage_status	false
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_T002_IN, PIPE_T002_OUT (mock IDs)
relationships.related_component_ids	V003, V004, V005 (mock IDs)